

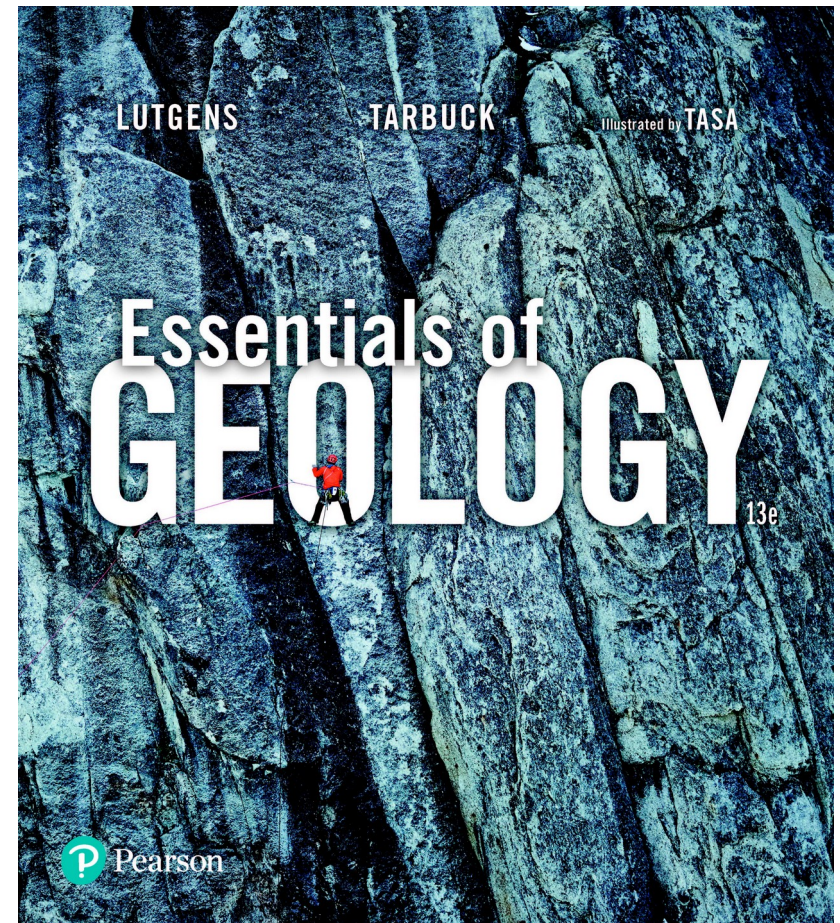
Lecture 8 Outline

Metamorphism & Metamorphic Rocks

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Essentials of Geology

Thirteenth Edition



Learning Objectives

- **LO 8.1** Compare and contrast the environments that produce metamorphic, sedimentary, and igneous rocks.
- **LO 8.2** List and distinguish among the four agents that drive metamorphism.
- **LO 8.3** Explain how foliated and nonfoliated textures develop.
- **LO 8.4** List and describe the most common metamorphic rocks.

Learning Objectives (cont.)

- **LO 8.5** Write a description for each of these metamorphic environments: contact metamorphism, hydrothermal metamorphism, subduction zone metamorphism, and regional metamorphism.
- **LO 8.6** Explain how index minerals are used to establish the metamorphic grade of a rock body.

8.1 What Is Metamorphism?

- Metamorphism means to “change form”
 - The transition of one rock into another by temperatures or pressures unlike those in which it formed
 - Changes in mineralogy and sometimes chemical composition
- Every metamorphic rock has a parent rock (the rock from which it formed).
 - Parent rocks can be igneous, sedimentary, or other metamorphic rocks.

8.1 What Is Metamorphism? (cont.)

- Metamorphic grade is the degree to which the parent rock changes during metamorphism.
 - Progresses from low grade (low temperatures and pressures) to high grade (high temperatures and pressures)
- During metamorphism, the rock must remain essentially solid.

Metamorphic Grade

A. Parent rock
(Shale)



Low-grade
metamorphism
Low temperatures
and pressures

Metamorphic rock
(Slate)



B. Parent rock
(Granodiorite)



High-grade
metamorphism

Strong
compressional
forces, high
temperatures
and pressures

Metamorphic rock
(Folded gneiss)



8.2 What Drives Metamorphism?

- Heat
 - Most important agent
 - Recrystallization is the process of forming new, stable minerals larger than the original.
 - Two sources of heat:
 - Geothermal gradient: An increase in temperature with depth (about 25°C per kilometer)
 - Contact metamorphism: Rising mantle plumes

8.2 What Drives Metamorphism? (cont.)

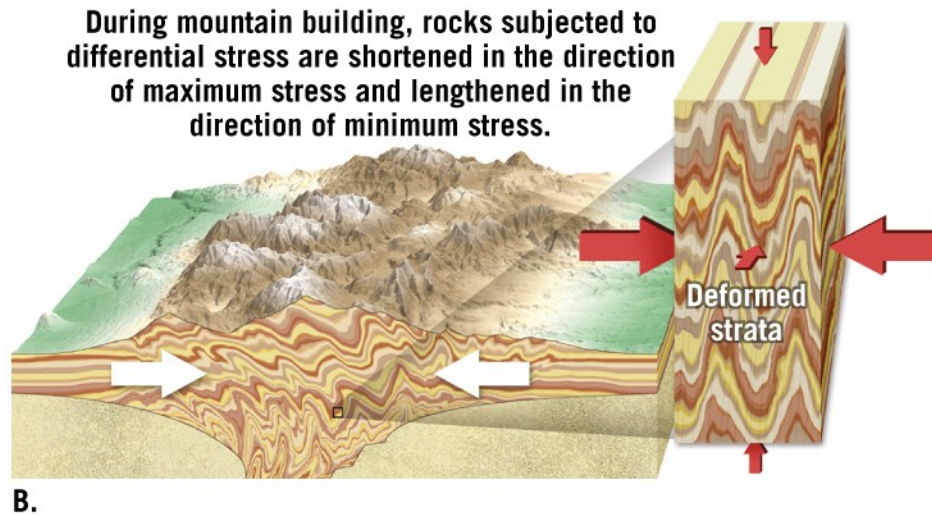
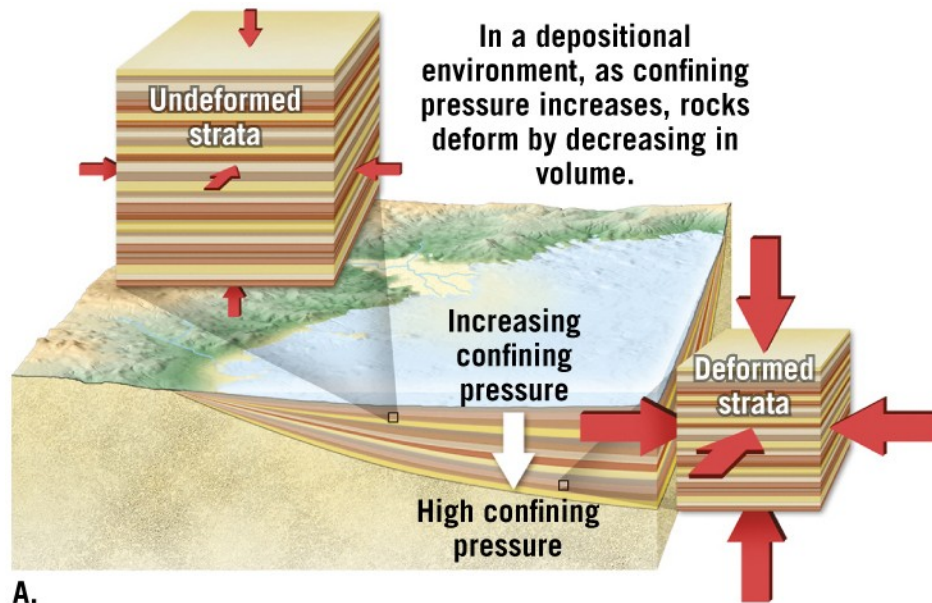
- Confining pressure
 - Forces are applied equally in all directions
 - Analogous to water pressure
 - Causes the spaces between mineral grains to close

8.2 What Drives Metamorphism? (cont.)

- Differential stress
 - Forces are unequal in different directions.
 - Stresses are greater in one direction
- Compressional stress
 - Rocks are squeezed as if in a vice.
 - Shortened in one direction and elongated in the other direction

Confining Pressure and Differential Stress

Tutorial



8.2 What Drives Metamorphism?

- Chemically active fluids
 - Enhances migration of ions
 - Aids in recrystallization of existing minerals
 - In some environments, fluids can transport mineral matter over considerable distances.

8.2 What Drives Metamorphism? (cont.)

- The importance of parent rock
 - Most metamorphic rocks have the same overall chemical composition as the original parent rock.
 - Mineral makeup determines the degree to which each metamorphic agent will cause change.

8.3 Metamorphic Textures

- Texture describes the size, shape, and arrangement of mineral grains.
 - Metamorphic rocks can display preferred orientation of minerals, in which the platy mineral grains exhibit parallel to subparallel alignment.

8.3 Metamorphic Textures (cont.)

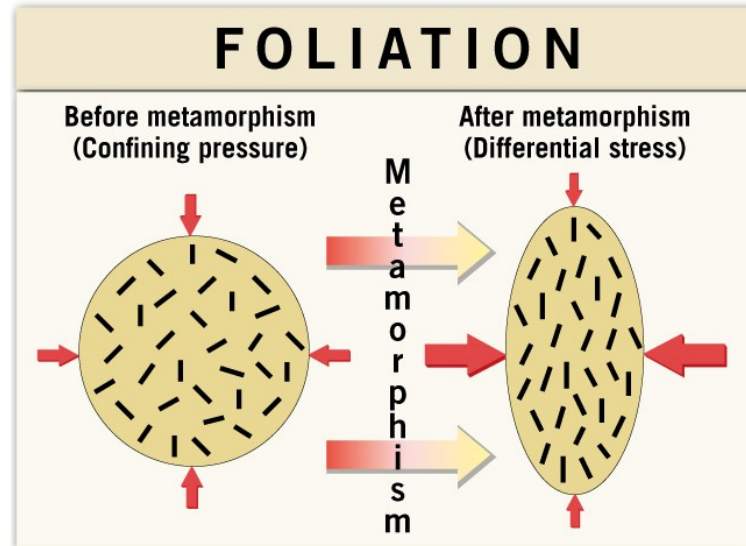
- Foliation describes any planar arrangement of mineral grains or structural features within a rock.
 - Examples of foliation
 - Parallel alignment of platy or elongated minerals
 - Parallel alignment of flattened mineral grains or pebbles
 - Compositional banding of dark and light minerals
 - Cleavage where rocks can be easily split into slabs

8.3 Metamorphic Textures (cont.)

- Foliation
 - Foliation can form in various ways, including:
 - Rotation of platy minerals
 - Recrystallization that produces new minerals perpendicular to the direction of maximum stress
 - Flattening spherically shaped grains

Development of Foliation as a Result of Differential Stress

[Animation](#)

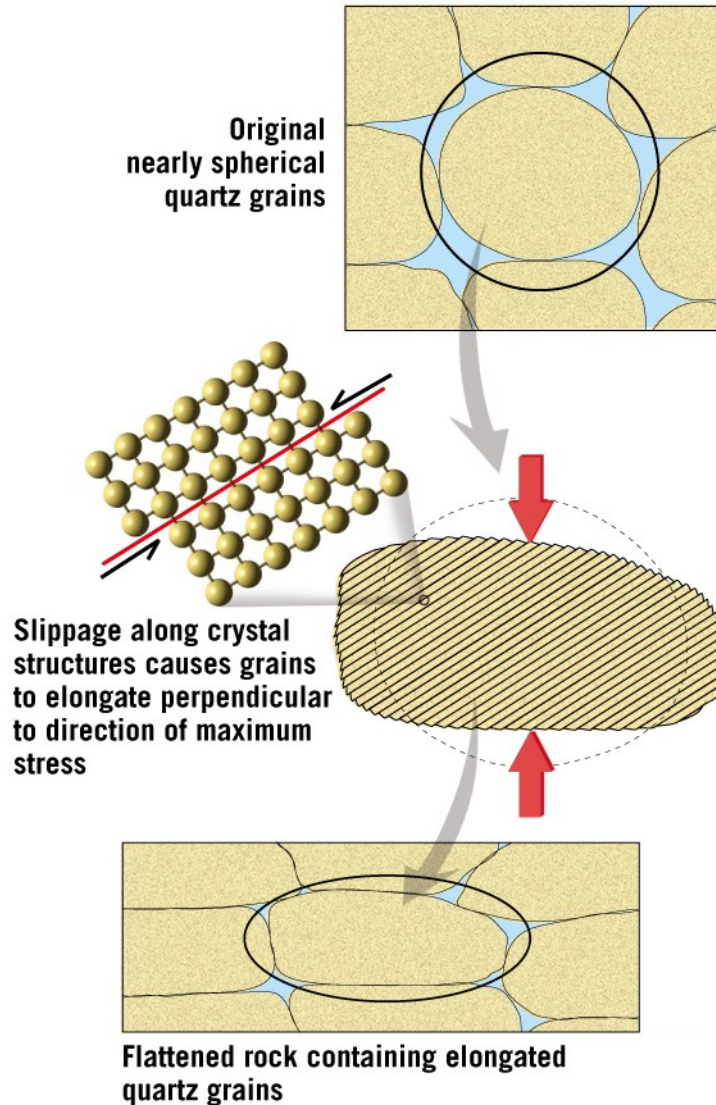


Platy and elongated mineral grains having random orientation.



When differential stress causes rocks to flatten, the mineral grains rotate and align roughly perpendicular to the direction of maximum differential stress.

Solid-State Flow of Mineral Grains



8.3 Metamorphic Textures

- Foliated textures
 - Rock or slaty cleavage
 - Rocks split into thin slabs
 - Develops in beds of shale with low-grade metamorphism

Excellent Slaty Cleavage



8.3 Metamorphic Textures

- Foliated textures
 - Schistosity
 - Platy minerals are discernible with the unaided eye.
 - Mica and chlorite flakes begin to recrystallize into large muscovite and biotite crystals.
 - Exhibit a planar or layered structure
 - Rocks having this texture are referred to as *schist*.

8.3 Metamorphic Textures (cont.)

- Foliated textures
 - Gneissic texture
 - During high-grade metamorphism, ion migration results in the segregation of minerals into light and dark bands.
 - Metamorphic rocks with this texture are called *gneiss*.
 - Although foliated, gneiss do not usually split as easily as slates and schists.

8.3 Metamorphic Textures (cont.)

- Other metamorphic textures
 - Nonfoliated metamorphic rocks are composed of minerals that exhibit equidimensional crystals and lack foliation.
 - Develop in environments in which deformation is minimal
 - Porphyroblastic textures
 - Unusually large grains, called porphyroblasts, are surrounded by a fine-grained matrix of other minerals.

Garnet-Mica Schist



Porphyroblasts



**Close up of
porphyroblast**

8.4 Common Metamorphic Rocks

Metamorphic Rock		Texture	Comments	Parent Rock
Slate		 Foliated	Composed of tiny chlorite and mica flakes, breaks in flat slabs called slaty cleavage, smooth dull surfaces	Shale, mudstone, or siltstone
Phyllite			Fine-grained, glossy sheen, breaks along wavy surfaces	Shale, mudstone, or siltstone
Schist			Medium- to coarse-grained, scaly foliation, micas dominate	Shale, mudstone, or siltstone
Gneiss			Coarse-grained, compositional banding due to segregation of light and dark colored minerals	Shale, granite, or volcanic rocks
Marble		 Non-foliated	Medium- to coarse-grained, relatively soft (3 on the Mohs scale), interlocking calcite or dolomite grains	Limestone, dolostone
Quartzite			Medium- to coarse-grained, very hard, massive, fused quartz grains	Quartz sandstone
Hornfels			Very fine-grained, often exceedingly tough and durable, usually dark colored	Often shale, but can have any composition

8.4 Common Metamorphic Rocks (cont.)

- Foliated rocks
 - Slate
 - Very fine-grained
 - Excellent rock cleavage
 - Most often generated from low-grade metamorphism of shale, mudstone, or siltstone

Slate



8.4 Common Metamorphic Rocks

- Foliated rocks
 - Phyllite
 - Degree of metamorphism between slate and schist
 - Platy minerals are larger than slate but not large enough to see with the unaided eye.
 - Glossy sheen and wavy surfaces
 - Exhibits rock cleavage

Phyllite



8.4 Common Metamorphic Rocks

- Foliated rocks
 - Schist
 - Medium- to coarse-grained
 - Parent rock is shale that has undergone medium- to high-grade metamorphism.
 - The term *schist* describes the texture.
 - Platy minerals (mainly micas) predominate
 - Can also contain porphyroblasts

Schist



8.4 Common Metamorphic Rocks

- Foliated rocks
 - Gneiss
 - Medium- to coarse-grained metamorphic rock with a banded appearance
 - The result of high-grade metamorphism
 - Composed of light-colored, feldspar-rich layers with bands of dark ferromagnesian minerals

Gneiss



8.4 Common Metamorphic Rocks

- Nonfoliated rocks
 - Marble
 - Crystalline rock from limestone or dolostone parent rock
 - Main mineral is calcite
 - Calcite is relatively soft (3 on the Mohs scale)
 - Used as a decorative and monument stone
 - Impurities in the parent rocks provide a variety of colors.

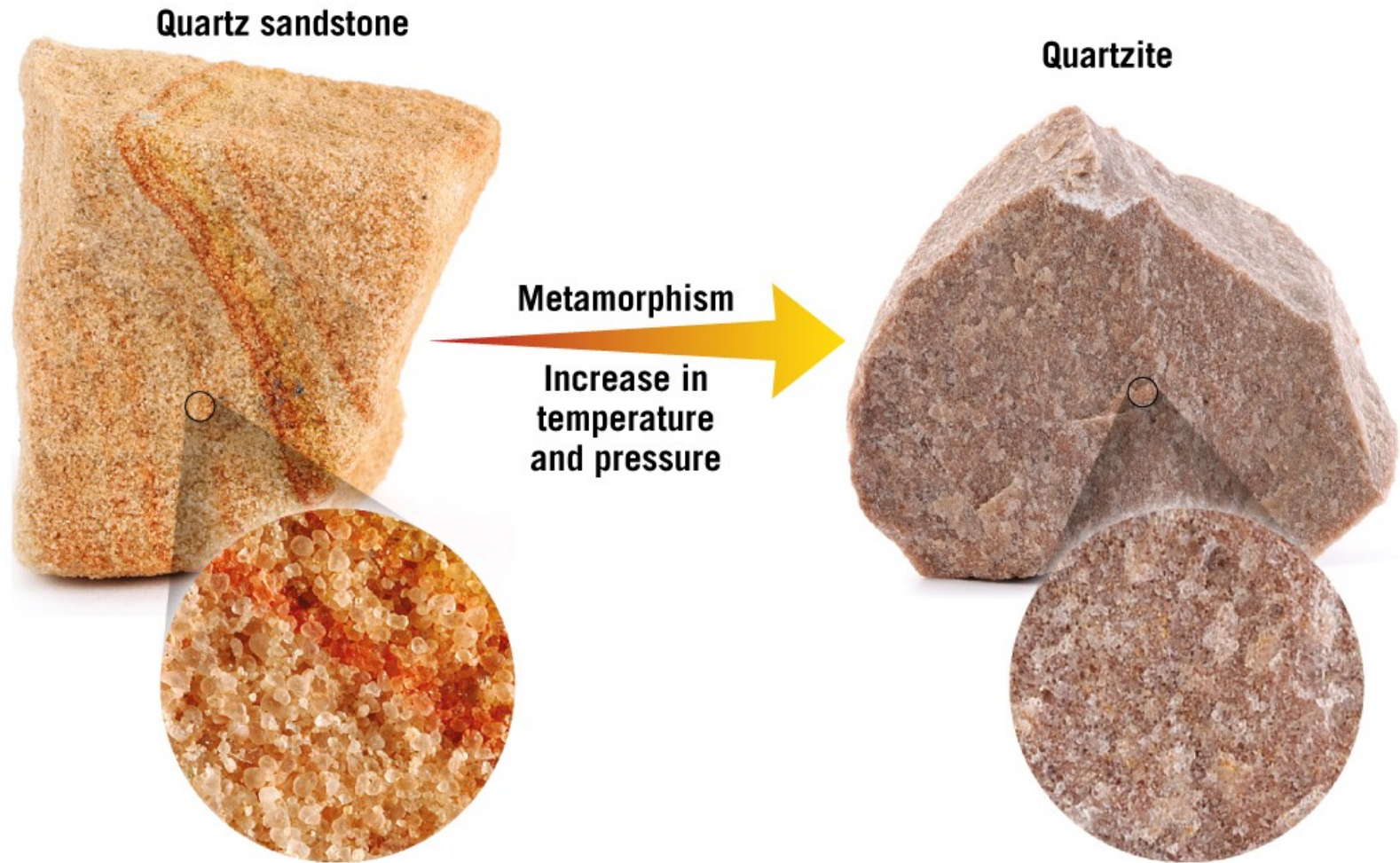
Marble



8.4 Common Metamorphic Rocks

- Nonfoliated rocks
 - Quartzite
 - Formed from a parent rock of quartz-rich sandstone
 - Quartz grains are fused together
 - Pure quartzite is white
 - Iron oxide may produce reddish or pink stains.
 - Dark minerals may produce green or gray stains.

Quartzite



8.4 Common Metamorphic Rocks

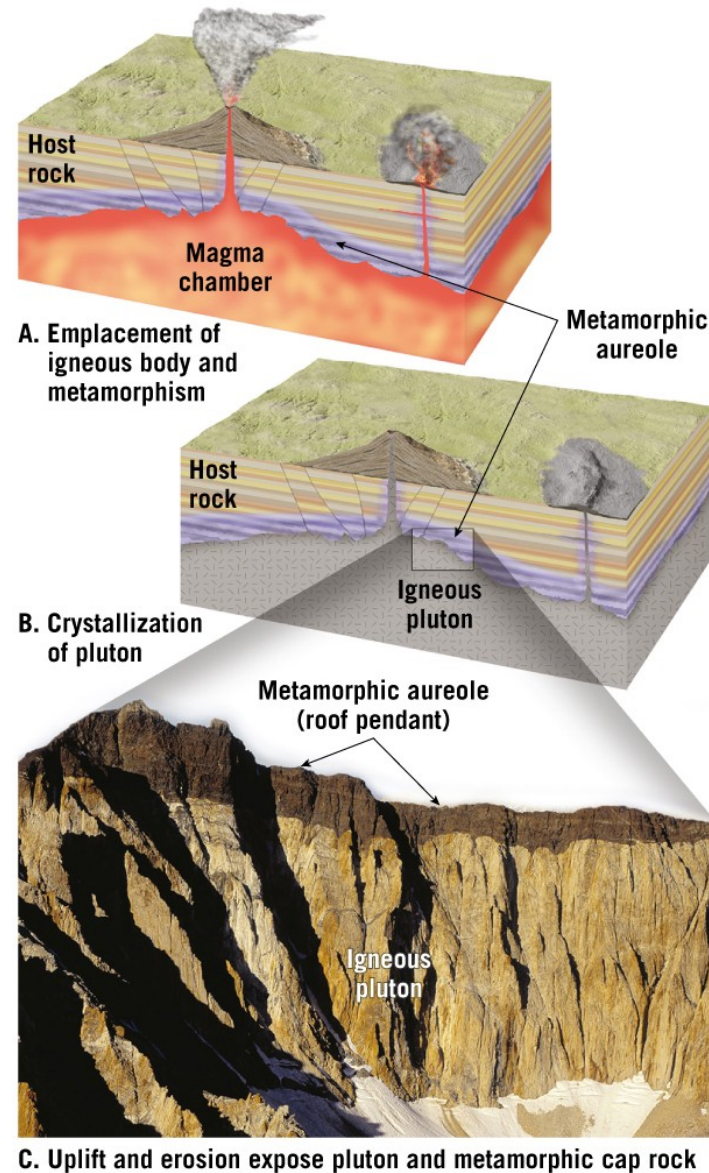
- Nonfoliated rocks
 - Hornfels
 - Parent rock is shale or clay-rich rocks.
 - “Baked” by an intruding magma body

8.5 Metamorphic Environments

- Contact or thermal metamorphism
 - Results from a rise in temperature when magma invades a host rock
 - Occurs in the upper crust (low pressure, high temperature)
 - The zone of alteration (aureole) forms in the rock surrounding the magma.

Contact Metamorphism

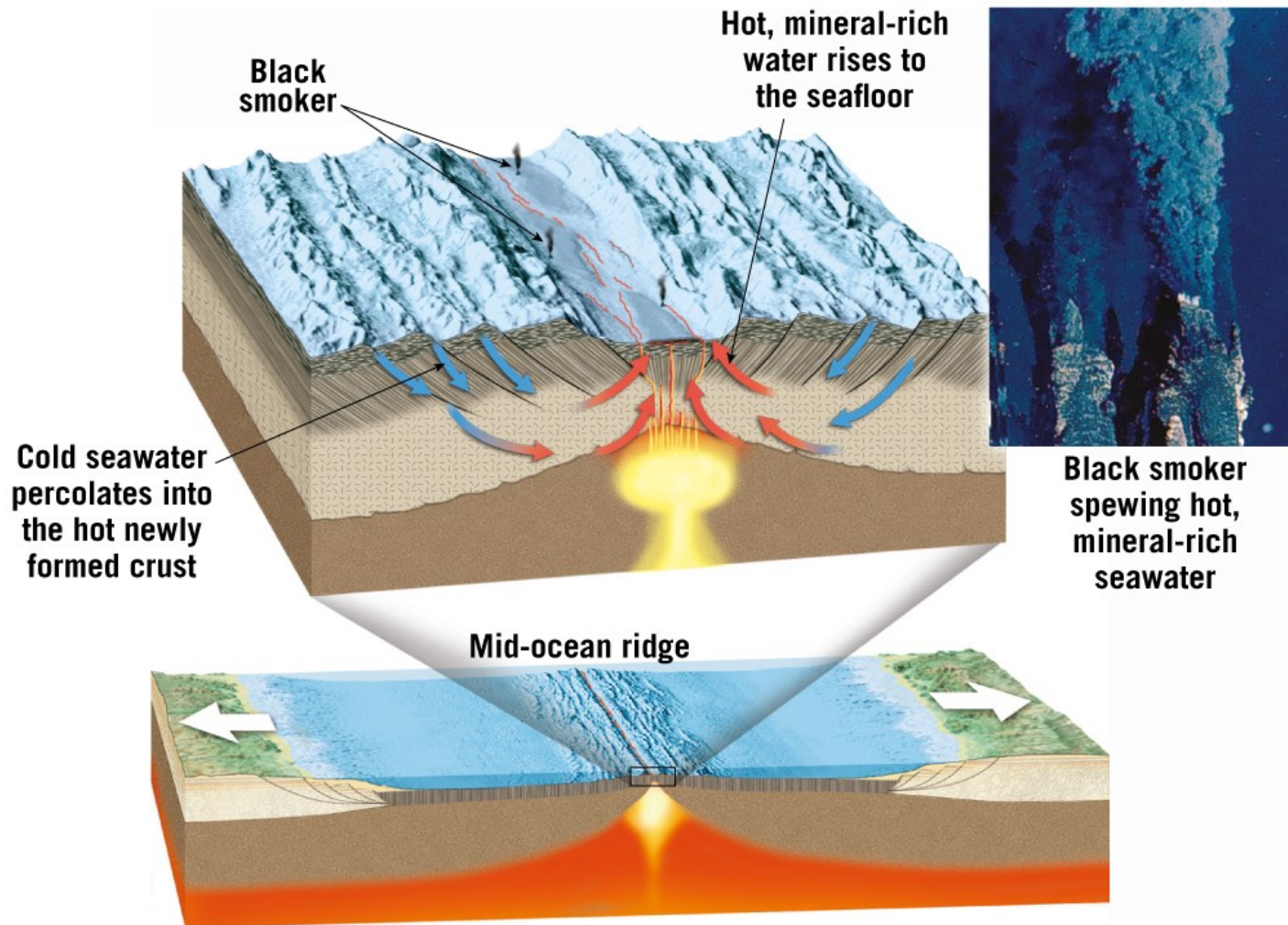
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8.5 Metamorphic Environments

- Hydrothermal metamorphism
 - Chemical alteration caused by hot, ion-rich fluids circulating through pore spaces and rock fractures.
 - Typically occurs along the axes of mid-ocean ridges.
 - Black smokers are the result of the fluids gushing from the seafloor.

Hydrothermal Metamorphism along a Mid-Ocean Ridge



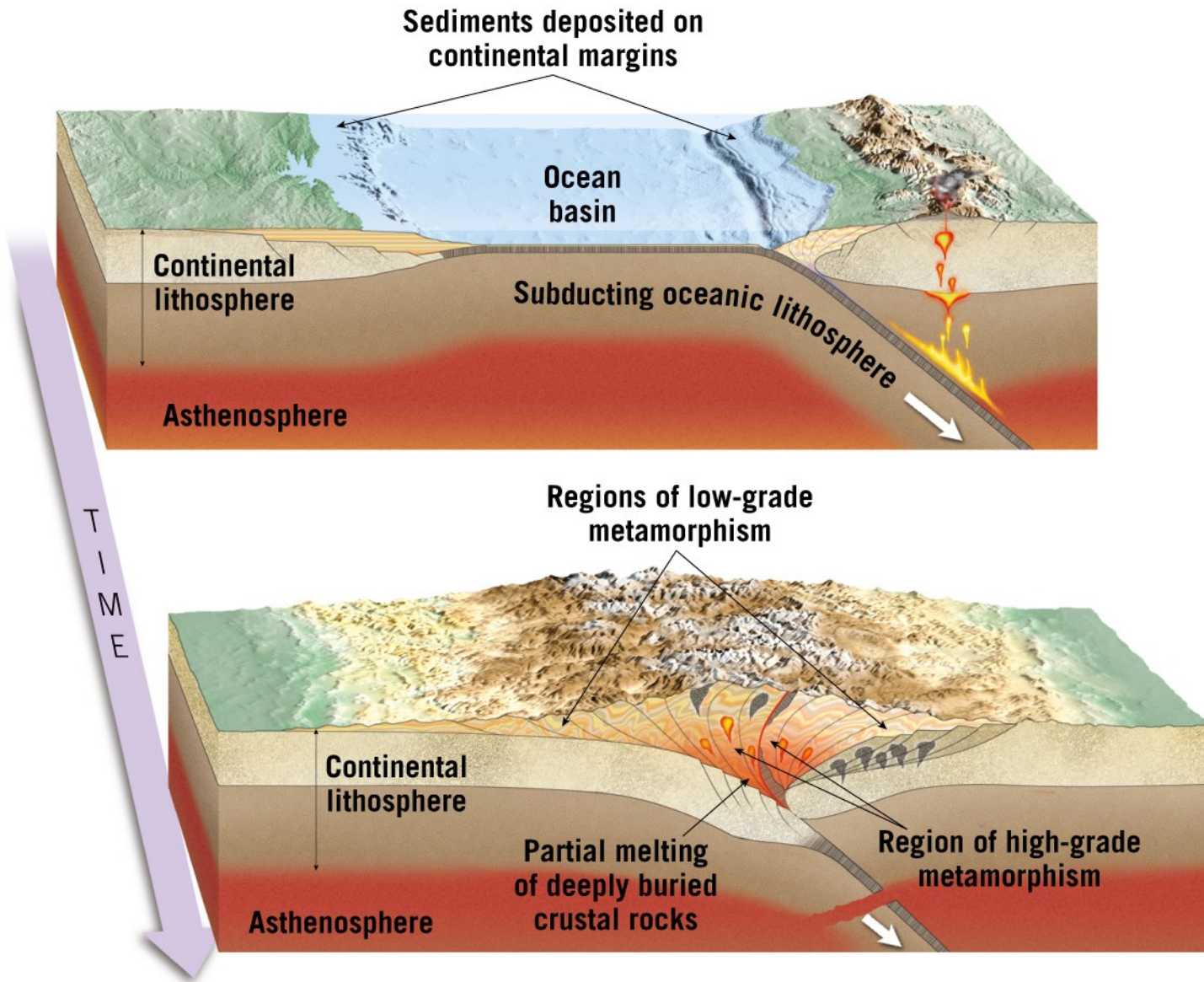
8.5 Metamorphic Environments

- Burial metamorphism
 - Associated with very thick sedimentary strata in a subsiding basin
 - Gulf of Mexico is an example.
- Subduction zone metamorphism
 - Sediments and oceanic crust are subducted fast enough that pressure increases before temperature.

8.5 Metamorphic Environments (cont.)

- Regional metamorphism
 - Produces the greatest quantity of metamorphic rock
 - Associated with mountain building and the collision of continental blocks

Regional Metamorphism

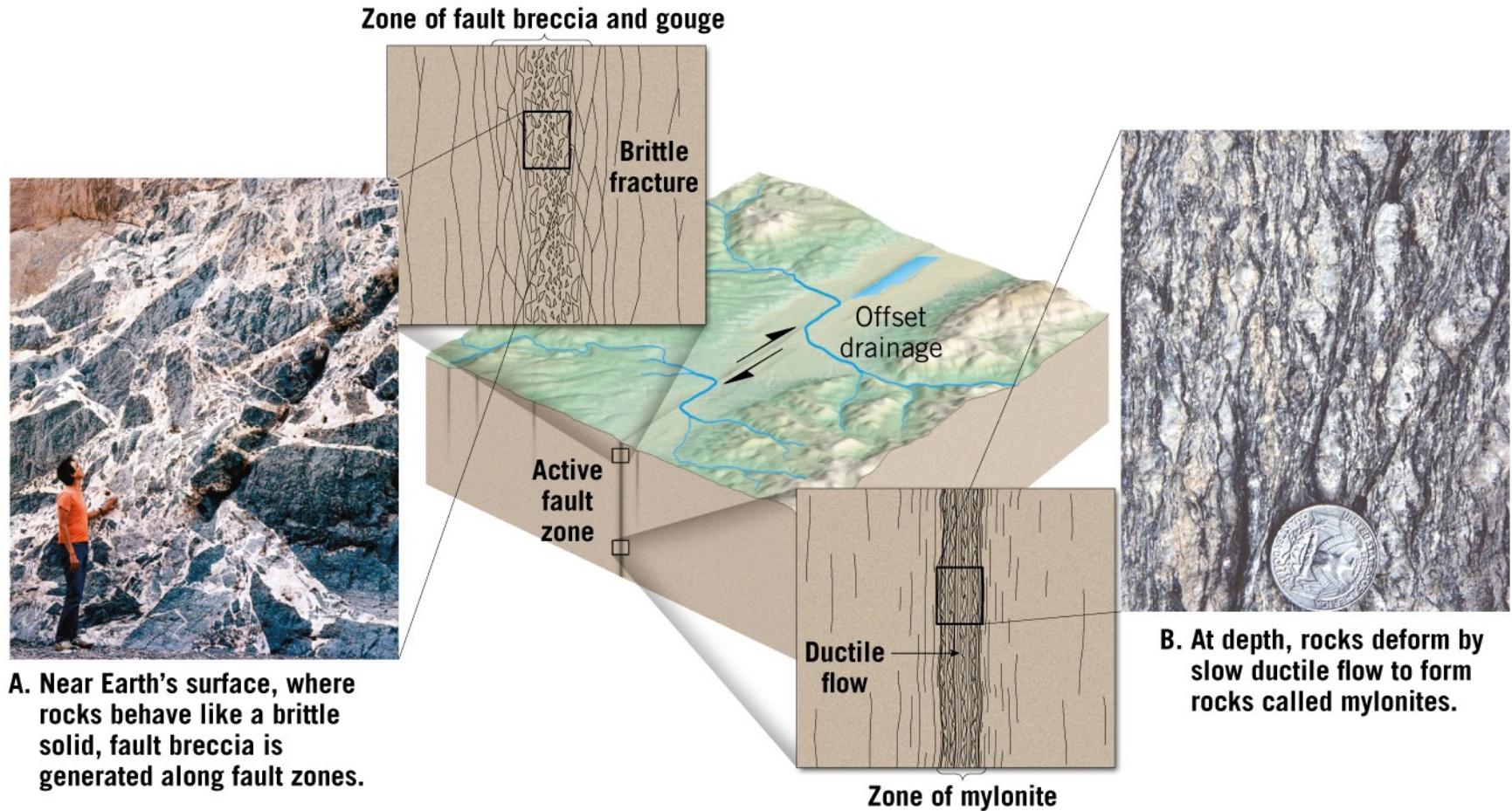


8.5 Metamorphic Environments

- Metamorphism along fault zones
 - Occurs at depth and high temperatures
 - Preexisting minerals deform by ductile flow
 - Mylonites form in these regions of ductile deformation.
- Impact metamorphism
 - Occurs when meteorites strike Earth's surface
 - Products of these impacts are fused fragmented rock and glass-rich ejecta that resemble volcanic bombs, called impactites.

Metamorphism along a Fault Zone

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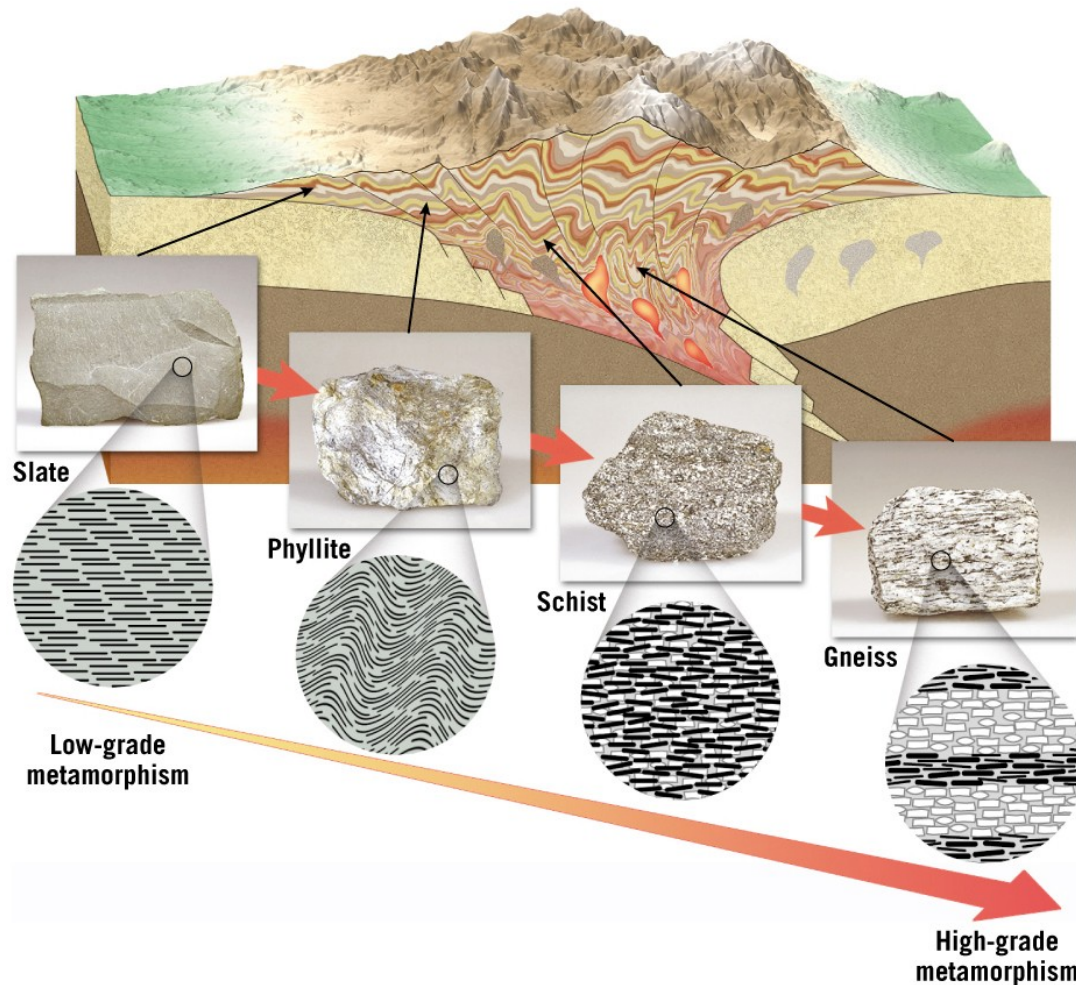


8.6 Metamorphic Zones

- Textural variations
 - Slate is associated with low-grade metamorphism.
 - Gneiss is associated with high-grade metamorphism.
 - Phyllite and schist are intermediate.

Textural Variations Caused by Regional Metamorphism

Tutorial



8.6 Metamorphic Zones

- Index minerals and metamorphic grade
 - Changes in mineralogy occur from regions of low-grade metamorphism to regions of high-grade metamorphism.
 - Index minerals are good indicators of metamorphic environments
 - Migmatites are rocks that are partially melted.
 - Represent the highest grades of metamorphism
 - Transitional to igneous rocks

Metamorphic Zones and Index Minerals

